

COP 701: Assignment 2

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Abstract

A 2D action-adventure game called METAL SLUG 6 is developed with Unity. We have used C# as the programming language for the development of the game. Using a variety of weaponry the player's character must destroy foes that randomly spawn, fire at random intervals, save itself from traps and obstacles and collect pick-up health power ups. With its own interactions and effects on foes, the game's mechanics include shooting, physical fighting, and explosive grenades. The aim is to endure for the longest duration possible and kill enemies to complete the mission that consists of three levels.

1 Introduction

Players must use a variety of tools and skills to fend off waves of soldeirs in this 2D action-adventure game. Players take control of a character that can shoot bullets, throw grenades, and engage in close quarters combat with a dagger in this dynamic setting. Every weapon has a different mechanism: the dagger kills quickly and precisely, grenades give area-of-effect damage, and bullets are used for ranged strikes in horizontal and vertical directions. Every confrontation is unexpected since enemies appear at key spots and fire at random times. Using the appropriate weapon in each situation, the player must carefully balance offence and defence as they progress through the game. The player's vitality is monitored via a health system, and proficient gameplay is rewarded with a scoring system that carries over points across stages. Because players must control weapon cooldowns, avoid opposing attacks, and tactically remove threats, the game places a strong emphasis on tactical decision-making. The immersive experience is enhanced with visual and aural feedback, such as animations and sound effects brought on by collisions and attacks. The game is tough and captivating, putting the player's reflexes and strategic thinking to the test as it gets harder.

2 Game Design

2.1 Player Mechanics

Mechanics for the game

- **Movement:** The player can move left and right using the left arrow and the right arrow keys, allowing for quick navigation across the screen.
- **Shooting:**
 - **Horizontal Shooting:** The player can shoot horizontally by pressing the S key.
 - **Vertical Shooting:** The player can shoot vertically using the W key, allowing for combat with bombs dropping from above.
- **Grenades:** By pressing the G key, the player tosses a grenade, which explodes and deals area-of-effect damage to enemies within a 1.5-unit radius.
- **Dagger:** Pressing the E key activates the dagger, instantly killing any enemy within 1.5 units of distance in front of the player.

- **Crouch:** The player can crouch by pressing C, lowering their height to avoid certain projectiles or obstacles. It can't shoot upward in crouch position. If W is pressed it will switch to standing position and then shoot. Or if space is pressed, it will jump and switch to standing state.
- **Jump:** The player jumps by pressing the space key, allowing them to traverse platforms or dodge enemy fire.

2.2 Enemy Design

2.2.1 Soldier Enemies

- These enemies approach the player and shoot bullets with rifles. They spawn at a certain distance from the player and have a basic movement pattern of advancing toward the player while firing.
- Soldier enemies are destroyed after being hit by 3 shots or killed by a grenade.

2.2.2 Stabbing Enemies

- These enemies approach the player and use axes to kill. They spawn at a certain distance from the player and have a basic movement pattern of advancing toward the player.
- Stabbing enemies are destroyed after being hit by 3 shots or killed by a grenade.

2.2.3 Bosses

Each level features a unique boss:

- **Level 1:** Shooting Truck
- **Level 2:** Giant Hermit
- **Level 3:** Tank

Bosses shoot cannons instead of using rifles, causing more significant damage to the player (reducing health by 3 units per hit). Bosses have much higher health points and require more hits to be destroyed.

2.3 Hazards

- **Bombs:** Bombs are dropped from the sky. By getting hit by it health get reduced. So player has to dodged it or use vertical bullets.
- **Electric traps:** Electric traps are present on the ground. By getting hit by it health get reduced. So player has to jump over it.

2.4 Obstacles

- **Level 2 (Arctic):** Sandbags are placed throughout the battlefield, serving as cover from enemy fire.
- **Level 3 (Nuclear Reactor):** Boxes are placed on the field, which block the player's and enemies' movement and attacks.

2.5 Hidden Power-ups

There are some hidden boxes present in the levels. Within those boxes health power-up is present, which can be gained by destroying the boxes. To make it more complicated in the third level we have same type of boxes, some are destructible and some are not.

2.6 Level Design

- **Level 1 (Jungle):** Enemies spawn at a moderate rate, and the player faces the Shooting Truck as the final boss. Traps and power-ups are present.
- **Level 2 (Arctic):** The spawning rate of enemies increases, and obstacles like sandbags appear on the battlefield. Traps are also present.
- **Level 3 (Nuclear Reactor):** The most challenging level. It has obstacles, traps, power-ups and the enemies spawn at a higher rate. The boss has much higher shot taken capabilities.

2.7 UI and Display

The top of the screen features key information:

- **Score:** The player's score is displayed, which increases with each enemy or boss kill. The score got carry forward by promoting to each level as the game is a total mission and it has 3 levels in it.
- **Weapons:** Shows the current weapons that the player has.
- **Health Bar:** The player's health is displayed, starting at full health and decreasing with each hit. At the beginning of each level the player health get to the full health.
- **Level Name:** The current level is displayed at the top.
- **Pause Button:** A pause button is also present below the level name. It is used to pause the game. By clicking it we will get a screen with 3 options - Main Menu, Restart and Resume.

2.8 Enemy Spawning

The rate at which enemies spawn increases with each level, adding to the game's difficulty. Soldier enemies will always spawn at a certain distance from the player, giving them time to engage before being overwhelmed.

2.9 Scoring and Damage

- **Score:**
 - The player earns points for every enemy they shoot or kill using grenades.
 - The score increases significantly when bosses are defeated.
- **Damage:**
 - Soldier enemies reduce the player's health by 1 unit each time they shoot and hit the player.
 - Bosses deal 3 units of damage to the player when their cannon shots hit.
 - Grenades cause damage equal to 5 shots when they hit any enemy or boss.
 - Player causes damage of 1 with bullets.
 - Traps reduces the health of the player by 10 units.
 - Bombs coming from the sky can damage the health by 5.

2.10 Health and Difficulty:

Player Health: Player has a health level of 100 at each level. **Enemy Health:**

- Soldier enemies require 3 shots to be destroyed.
- Level 1 enemies take 300 shots.
- Level 2 enemies take 500 shots.

- Level 3 enemies take 750 shots to be killed.
- Grenades: Each grenade deals damage equivalent to 5 shots, making them powerful tools against tougher enemies and bosses.

This game combines strategic shooting, diverse weapon use, and varied enemy behavior, increasing the challenge as the player progresses through three distinct levels with increasing difficulty. The player must balance their arsenal and movement to survive waves of enemies, traps and obstacles and defeat powerful bosses.

3 Challenges

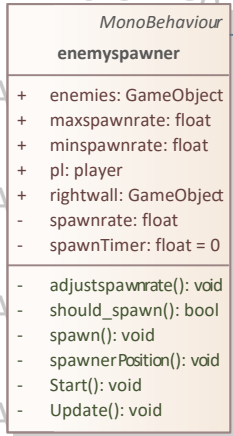
- Synchronize the animations of upper body and lower body of the main player
- Synchronize the instantiating of the bullets, dagger and cannons with animations
- Carrying score forward in next level
- Synchronizing overall animations with gamellogic

4 References

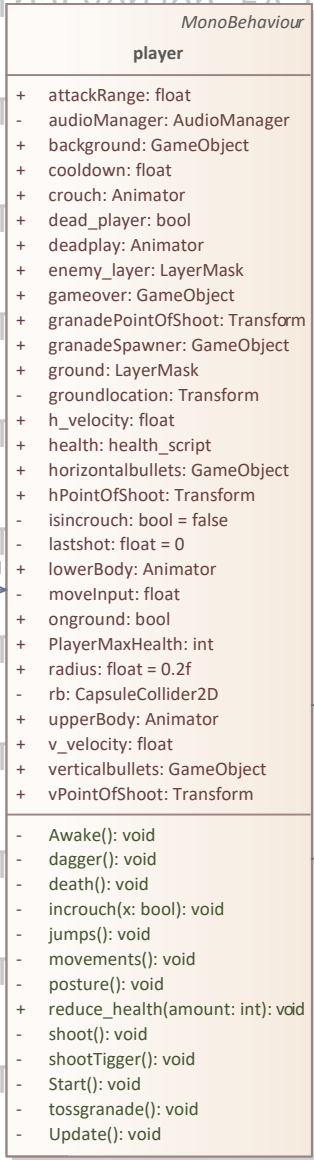
1. [Making 2D game in unity](#)
2. [Flappy bird game](#)
3. [Audio Files](#)
4. [Metal Slug Sprites-1](#)
5. [Metal Slug Sprites-2](#)

5 Implementation

The implementation of the game logic is presented in the UML diagram from the next page.



+pl



+health

